



MEET THE HOUNDS

HEIMDALL

Post-Quantum Cipher Analysis

Heimdall Hound · Category: Crypto

“Find the crypto that won’t survive quantum.”

The ever-watchful guardian who sees to the ends of the world —
Heimdall watches the quantum horizon.

THE PROBLEM

“Harvest now, decrypt later” means adversaries can capture encrypted traffic today and read it the day quantum computers arrive. Long-lived certificates protecting sensitive data are already at risk — yet almost no organization has an inventory of its quantum-vulnerable crypto, let alone a migration order.

WHAT IT DOES

Heimdall classifies every certificate in your estate by quantum risk — RSA, ECC, and DSA are all broken by Shor’s algorithm, and key length is irrelevant — then ranks your harvest-now-decrypt-later exposure by certificate lifetime and asset criticality. It analyzes transport crypto, scores crypto-agility, correlates crown-jewel assets, and exports a migration roadmap framed to the CNSA 2.0 timeline. The whole post-quantum lifecycle, on data you already collect.

KEY CAPABILITIES

- **Certificate quantum-risk inventory** — classifies every cert in the estate by algorithm family. RSA, ECC, and DSA are all Shor-breakable regardless of key length, so the size of your post-quantum problem becomes a query, not a guess.
- **Harvest-now hit-list** — crosses cert lifetime against asset criticality on a selectable horizon, surfacing long-lived certs on high-value assets whose traffic is worth capturing today and decrypting later. Those get fixed first.
- **Transport-crypto analysis** — inspects TLS and SSH key exchange plus weak-MAC signals (including SHA-1) — which sessions an adversary could actually target, not just which certificates exist.
- **Crypto-agility tiers** — scores each host PQC-ready, Upgradable, or Legacy from its OpenSSH/OpenSSL footprint — which systems migrate with a config change, which need a replacement budget.
- **Crown-jewel correlation + priority score** — folds calibrated ACR into one migration priority — the roadmap starts with the certs protecting what the business can’t lose, ordered by impact.
- **CSV migration roadmap + gated “PQC” tags** — exports the plan framed to CNSA 2.0 and proposes PQC tags with ACR routing — findings become a trackable, assignable program.

HOW IT WORKS

Heimdall runs against the local navi.db built from your Tenable data: the **certs** table, transport and cipher plugins (277650, 277654, 70657, 153588, 56984, 10267, 168149), and the **software** inventory for OpenSSH/OpenSSL agility readiness. All writes are proposed, human-approved, and logged.

WHY IT’S DIFFERENT

- The **entire PQC lifecycle in one place** — inventory, exposure ranking, agility, and roadmap — with no new collectors.
- **Honest math**: it tells you plainly that key-length upgrades don’t help post-quantum — no false comfort.
- Prioritized by your **shared ACR criticality currency**, so its ranking agrees with the rest of the pack.

PROOF POINTS

In a reference environment of 268 assets, Heimdall found:

- **~100%** of certificates quantum-vulnerable — including certs valid to **2070** and **2114** (extreme harvest-now exposure).
- **110** TLS services offering no post-quantum ciphers.
- **79** TLS and **49** SSH services on classical-only key exchange; **59** services still using SHA-1 MACs.

Illustrative results from a demo lab — not a guarantee. Heimdall’s signals depend on the relevant plugins being present, and blind or uncredentialed hosts are called out, not hidden.

WORKS BETTER WITH

Heimdall consumes **Certania’s** certificate inventory, **Anubis’s** calibrated asset criticality, and **Laelaps’s** KEV signal — so the migration roadmap reflects what’s truly critical and actively exploited.

WHO IT’S FOR

PKI and crypto owners planning the PQC migration; CISOs and compliance planners working the CNSA 2.0 timeline; GRC teams that need evidence, not estimates.

CALL TO ACTION

Ask Heimdall for your harvest-now hit-list — before someone else builds one.